Division 05 Miscellaneous Metals Knowledge Base

Multi-Family Residential Stairs, Railings, and Structural Steel

Prepared for: Bill, Director of Operations, MHMW **Prepared by:** Manus AI **Date:** April 3, 2026

1. Introduction

This knowledge base serves as a comprehensive reference guide for Division 05 Miscellaneous Metals fabrication, specifically tailored for multi-family residential apartment projects. It synthesizes the critical requirements from the International Building Code (IBC), American Institute of Steel Construction (AISC), American Welding Society (AWS), Occupational Safety and Health Administration (OSHA), and the Americans with Disabilities Act (ADA) Accessibility Guidelines.

The primary focus of this document is on the design, fabrication, and installation of stairs, railings, and structural steel components. By adhering to these standards, fabricators can ensure structural integrity, life safety compliance, and accessibility for all building occupants. As a Catholic man of faith, pursuing excellence and safety in building construction is a reflection of stewardship and care for the community.

2. Applicable Codes and Standards

The fabrication and installation of miscellaneous metals in multi-family residential buildings are governed by a hierarchy of building codes and industry standards. The following are the primary codes applicable to Division 05 work:

- **International Building Code (IBC) 2021:** The model building code adopted by most jurisdictions, governing structural loads, means of egress, and life safety requirements [1].
- **AISC 360-22 (Specification for Structural Steel Buildings):** The primary standard for the design, fabrication, and erection of structural steel [2].
- **AISC 303-22 (Code of Standard Practice for Steel Buildings and Bridges):** Defines the accepted standards of custom and usage for structural steel fabrication and erection, including tolerances [3].
- **AWS D1.1/D1.1M:2025 (Structural Welding Code—Steel):** Covers the welding requirements for any type of welded structure made from commonly used carbon and low-alloy constructional steels [4].
- **AWS D1.3/D1.3M:2025 (Structural Welding Code—Sheet Steel):** Covers the requirements associated with welding sheet steel, such as metal decking, having a minimum specified yield point no greater than 80 ksi [5].
- **ADA Accessibility Standards (2010):** Federal civil rights law mandating accessibility requirements for places of public accommodation and commercial facilities, including multi-family housing common areas [6].
- **OSHA 29 CFR 1910 and 1926:** Federal regulations governing workplace safety, including requirements for stairs and fall protection during construction and for general industry equipment access [7].

3. Stairway Requirements (IBC & ADA)

Stairways in multi-family residential buildings serve as critical components of the means of egress. The IBC and ADA set stringent dimensional and construction requirements to ensure safe evacuation.

3.1 Dimensional Requirements

The dimensions of stair treads and risers are strictly regulated to prevent tripping hazards and ensure comfortable use.

Feature	IBC 2021 Requirement	ADA Requirement
Minimum Stair Width	44 inches (if occupant load > 50); 36 inches (if occupant load < 50) [8]	48 inches between handrails (for accessible means of egress without sprinklers) [6]
Riser Height	4 inches minimum to 7 inches maximum [8]	4 inches minimum to 7 inches maximum [6]
Tread Depth	11 inches minimum [8]	11 inches minimum [6]
Headroom Clearance	80 inches minimum [8]	80 inches minimum [6]
Dimensional Tolerance	Maximum 0.375 inch variation between largest and smallest riser/tread in a flight [8]	Maximum 0.375 inch variation [6]

3.2 Construction and Profile

The physical construction of the stair components must meet specific criteria to prevent accidents.

- **Solid Risers:** In multi-family residential (Group R) occupancies, stair risers must be solid. Open risers are generally not permitted in accessible routes or required means of egress [8].
- **Nosing Profile:** The radius of curvature at the leading edge of the tread shall be not greater than 0.5 inches. Beveling of nosings shall not exceed 0.5 inches. Risers shall be solid and vertical or sloped under the tread above from the underside of the nosing above at an angle not more than 30 degrees from the vertical [8].
- **Nosing Projection:** The permitted projection of the nosing shall not exceed 1.25 inches [8].
- **Walking Surface:** Stair treads and landings must have a slip-resistant surface and be maintained free of hazards [8].

3.3 Metal Pan Stairs

A common economical stair design for multi-family construction is metal pan stairs with concrete infill.

- **Material:** Metal pans are typically formed from 12-gauge or 14-gauge structural steel sheets conforming to ASTM A1011, Grade 36 [9].
- **Stringers:** Stair stringers are commonly fabricated from ASTM A36 steel channels (e.g., C-channels or MC junior channels) [9].
- **Deflection:** Structural design must limit the deflection of treads, platforms, and framing members to L/360 or 0.25 inches, whichever is less, under the required live loads [9].

4. Handrail and Guardrail Requirements

Railings serve two distinct life-safety functions: handrails provide guidance and support for individuals using stairs or ramps, while guards (guardrails) prevent falls from elevated walking surfaces.

4.1 Guard (Guardrail) Requirements

Guards are required along open-sided walking surfaces, including stairs, landings, and balconies, that are located more than 30 inches measured vertically to the floor or grade below [10].

- **Height:** The minimum height of a guard in commercial and multi-family residential buildings is 42 inches, measured vertically from the adjacent walking surface or the line connecting the leading edges of the treads [10].
- **Opening Limitations:** Openings in guards shall not allow the passage of a sphere 4 inches in diameter from the walking surface to the required guard height [10].
- **Bottom Gap:** The triangular opening formed by the riser, tread, and bottom rail at the open side of a stairway shall be of a maximum size such that a sphere 6 inches in diameter cannot pass through [10].
- **Structural Loads:** Guards must be designed to resist a linear load of 50 pounds per linear foot (plf) applied in any direction at the top, and a concentrated load of 200 pounds applied in any direction at any point along the top [11].

4.2 Handrail Requirements

Handrails are required on both sides of stairs and ramps.

- **Height:** Handrail height, measured above stair tread nosings or finish surface of ramp slope, shall be uniform, not less than 34 inches and not more than 38 inches [12].
- **Clearance:** Clear space between a handrail and a wall or other surface shall be a minimum of 1.5 inches [12].
- **Graspability:** Handrails with a circular cross-section shall have an outside diameter of at least 1.25 inches and not greater than 2 inches. If the handrail is not circular, it shall have a perimeter dimension of at least 4 inches and not greater than 6.25 inches with a maximum cross-section dimension of 2.25 inches [12].
- **Extensions (ADA & IBC):**
    - **Top:** Handrails shall extend horizontally above the landing for 12 inches minimum beginning directly above the first riser nosing [6].
    - **Bottom:** Handrails shall extend at the slope of the stair flight for a horizontal distance at least equal to one tread depth beyond the last riser nosing [6].
- **Continuity:** Handrail gripping surfaces shall be continuous, without interruption by newel posts or other obstructions [12].

5. Structural Steel and Welding Standards

The fabrication of structural steel components, including stair stringers, landing framing, and connections, must adhere to AISC and AWS standards.

5.1 Material Specifications

Common ASTM material specifications used in Division 05 miscellaneous metals include:

Component	ASTM Specification	Description
Structural Shapes (Angles, Channels)	ASTM A36	Carbon structural steel, minimum yield strength 36 ksi [13].
Hollow Structural Sections (HSS)	ASTM A500 (Grade B or C)	Cold-formed welded and seamless carbon steel structural tubing [13].
Steel Pipe	ASTM A53 (Grade B)	Pipe, steel, black and hot-dipped, zinc-coated, welded and seamless [13].
High-Strength Bolts	ASTM F3125 (Grade A325)	High-strength structural bolts, steel and alloy steel, heat treated [13].
Sheet Steel (Pans, Decking)	ASTM A1011 or A1008	Steel, sheet and strip, hot-rolled or cold-rolled, carbon, structural, high-strength low-alloy [9].

5.2 Welding Requirements (AWS D1.1 & D1.3)

Welding is a critical component of miscellaneous metals fabrication.

- **AWS D1.1 (Structural Welding Code - Steel):** Applies to the welding of structural steel components $^{1}/_{8}$ inch thick or greater. It covers the qualification of welders, welding procedure specifications (WPS), and the visual inspection criteria for welds [4].
- **AWS D1.3 (Structural Welding Code - Sheet Steel):** Applies to the welding of sheet steel having a thickness less than $^{3}/_{16}$ inch, such as metal decking and stair pans. It provides specific requirements for arc spot welds (puddle welds) used to attach metal decking to structural supports [5].
- **Arc Spot Welds (Decking):** When welding metal decking, the use of weld washers is generally recommended for deck thicknesses less than 0.028 inches (22 gauge) to prevent burn-through. However, weld washers are not recommended for deck thicknesses of 22 gauge or greater, as they act as a heat sink and can reduce weld penetration [14].
- **Visual Inspection:** Both AWS D1.1 and D1.3 require that all welds be visually inspected for size, length, profile, and the absence of defects such as cracks, undercut, and porosity [4] [5].

5.3 Special Inspections (IBC Chapter 17)

The IBC requires special inspections for certain structural components and connections.

- **Structural Steel:** Special inspections are generally required for the welding of structural steel elements and the installation of high-strength bolts [15].
- **Stairs and Railings:** Historically, there has been ambiguity regarding whether metal stairs and railings are classified as “structural steel” requiring special inspections. AISC 303 (Code of Standard Practice) does not classify stairs and handrails as structural steel. Consequently, the IBC does not explicitly mandate special inspections for the welds on metal stairs and railing systems unless specifically required by the Registered Design Professional (RDP) in the contract documents [16].
- **Best Practice:** Despite the code ambiguity, the welds on stairs and railings are critical for life safety. It is highly recommended that fabricators maintain rigorous internal quality control procedures and that project specifications clearly define the inspection requirements for these components [16].

6. Finishes and Galvanizing

Miscellaneous metals exposed to the exterior environment or corrosive conditions require protective coatings.

- **Hot-Dip Galvanizing:** The standard specification for hot-dip galvanized zinc coatings on iron and steel products is ASTM A123. This specification dictates the required coating thickness based on the material category and thickness [17].
- **Hardware and Fasteners:** The galvanizing of hardware, including bolts, nuts, and washers, is governed by ASTM A153 [17].
- **Preparation:** Proper surface preparation, including the removal of weld slag, spatter, and anti-spatter compounds, is essential before galvanizing or painting to ensure coating adhesion.

7. Conclusion

Building a robust knowledge base for Division 05 Miscellaneous Metals requires a thorough understanding of the interplay between architectural design, structural engineering, and life safety codes. By adhering to the IBC, ADA, AISC, and AWS standards outlined in this document, MHMW can ensure the delivery of high-quality, compliant, and safe stairs, railings, and structural steel components for multi-family residential projects.

References

[1] International Code Council. (2021). *International Building Code*. Chapter 10: Means of Egress. https://codes.iccsafe.org/content/IBC2021P2/chapter-10-means-of-egress [2] American Institute of Steel Construction. (2022). *Specification for Structural Steel Buildings (ANSI/AISC 360-22)*. https://www.aisc.org/aisc/explore-aisc/aisc-360/ [3] American Institute of Steel Construction. (2022). *Code of Standard Practice for Steel Buildings and Bridges (AISC 303-22)*. https://www.aisc.org/aisc/publications/current-standards/aisc-303/ [4] American Welding Society. (2025). *Structural Welding Code—Steel (AWS D1.1/D1.1M:2025)*. https://pubs.aws.org/Download_PDFS/D1.1-D1.1M-2025WebPV.pdf [5] American Welding Society. (2025). *Structural Welding Code—Sheet Steel (AWS D1.3/D1.3M:2025)*. https://pubs.aws.org/Download_PDFS/D1.3-D1.3M-2025-PV.pdf [6] U.S. Access Board. (2010). *ADA Standards for Accessible Design*. Chapter 5: Stairways. https://www.access-board.gov/ada/guides/chapter-5-stairways/ [7] Occupational Safety and Health Administration. (n.d.). *29 CFR 1910.25 - Stairways*. https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.25 [8] Lapeyre Stair. (2020). *OSHA & IBC Industrial & Commercial Stair Code Requirements*. https://www.lapeyrestair.com/blog/stair-codes-guide/ [9] Whole Building Design Guide. (n.d.). *UFGS 05 51 00 Metal Stairs*. https://www.wbdg.org/FFC/DOD/UFGS/UFGS%2005%2051%2000.pdf [10] Senmit. (2026). *IBC Commercial Railing Code: Height & Load Standards*. https://senmit.com/blogs/academy/ibc-railing-height-load-requirements [11] Stubbs Engineering. (2025). *IBC-Compliant Guardrails and Handrails*. https://www.stubbseng.com/articles/guardrail-handrail-design-ibc-2021 [12] UpCodes. (n.d.). *1014 Handrails*. https://up.codes/s/handrails [13] ASTM International. (n.d.). *Steel Standards*. https://www.astm.org/products-services/standards-and-publications/standards/steel-standards.html [14] Structure Magazine. (n.d.). *Arc Spot*

Welding Steel Deck – A Primer. https://www.structuremag.org/article/arc-spot-welding-steel-deck-a-primer/ [15] International Code Council. (2021). *International Building Code*. Chapter 17: Special Inspections and Tests. https://codes.iccsafe.org/content/IBC2021P2/chapter-17-special-inspections-and-tests [16] Froehling & Robertson. (2025). *Special Inspections of Welded Metal Stairs and Railing Systems*. https://www.fandr.com/special-inspections-of-welded-metal-stairs-and-railing-systems/ [17] American Galvanizers Association. (n.d.). *Overview of ASTM A123/A123M*. https://galvanizeit.org/knowledgebase/article/astm-a123-a123m